

★ Read the questions carefully
 ★ Make sure that you answered everything

Problem 1. Use the regression results below to answer the questions. $lprice$ log(price, \$1,000), $lsqrft$ log(size of house in square feet), $llotsize$ log(size of lot in square feet), *colonial* colonial style (=1), *bdrms* number of bedrooms

. reg lprice lsqrft llotsize colonial bdrms

Source	SS	df	MS	Number of obs	=	88
Model	5.20397919	4	1.3009948	F(4, 83)	=	38.38
Residual	2.81362433	83	.033899088	Prob > F	=	0.0000
				R-squared	=	0.6491
				Adj R-squared	=	0.6322
Total	8.01760352	87	.092156362	Root MSE	=	.18412

<u>lprice</u>	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
lsqrft	.7071931	.092802	7.62	0.000	.5226138	.8917725
llotsize	.1678189	.0381807	4.40	0.000	.0918791	.2437587
<u>colonial</u>	.0537962	.0447732	1.20	0.233	-.035256	.1428483
bdrms	.0268305	.0287236	0.93	0.353	-.0302995	.0839605
_cons	-1.349589	.651041	-2.07	0.041	-2.644483	-.0546947

(a) Interpret the coefficient on $lsqrft$

log-log A 1% increase in the size of the house increases the housing price by 0.71%, all else equal.

(b) Interpret the coefficient on *colonial*

log-lin Housing prices are 5.38% higher for colonial-style houses compared to non-colonial style houses, all else equal. However, the estimate is not statistically sig. at the 5% lv.

(c) Interpret the coefficient on *bdrms*

log-lin An add'l bedroom increases housing prices by 2.68%, all else equal. However, the estimate is not statistically sig. at the 5% lv.

Problem 2. Based on the table below, answer the questions.

Dependent Variable=Labor Force Participation				↓	↓
		(1)	(2)	(3)	(4)
		LPM	Probit	LPM	Probit
β_1	HHI	-0.11** (0.006)	-0.38** (0.019)	-0.09** (0.006)	-0.32** (0.021)
β_2	LF experience			-0.01** (0.000)	-0.03** (0.001)
β_3	nchild<6yrs old			-0.14** (0.005)	-0.49** (0.016)
β_4	nchild 6~18yrs old			-0.02** (0.003)	-0.09** (0.010)
β_5	Husband's earned income (\$10,000)			-0.01** (0.001)	-0.02** (0.004)
β_0	Constant	0.83** (0.004)	0.97** (0.016)	1.09** (0.008)	1.97** (0.039)
	Observations	22,369	22,369	22,259	22,259
	R-squared	0.017		0.071	

Robust standard errors in parentheses

** p<0.01, * p<0.05, + p<0.1

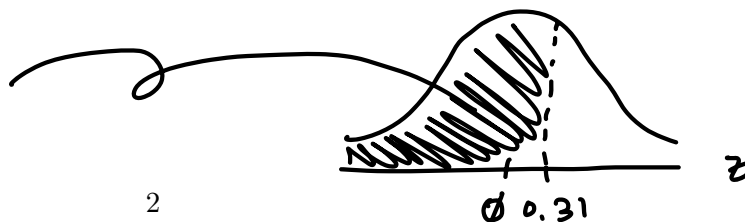
$$P(LFP=1) = \beta_0 + \beta_1 HHI + \beta_2 exp + \beta_3 nch6 + \beta_4 nch18 + \beta_5 hinc + u$$

- (a) Suppose a married woman is covered by her husband's insurance (HHI), has 10 years of working experience, and 2 children who are younger than 6 years old and her husband's income is \$30,000. In column (3), what is the probability that the woman will participate in the labor force?

$$\begin{aligned}
 P(LFP=1 \mid HHI=1, exp=10, nchild<6=2, nchild6-18=0, hinc=3) \\
 &= 1.09 + (-0.09) + (-0.01) \cdot 10 + (-0.14) \cdot 2 + 0 + (-0.01) \cdot 3 \\
 &= 0.59
 \end{aligned}$$

- (b) In column (4), what is the probability that the woman will participate in the labor force?

$$\begin{aligned}
 P(LFP=1 \mid HHI=1, exp=10, nchild<6=2, nchild6-18=0, hinc=3) \\
 &= \Phi(1.97 - 0.32 + (-0.03) \cdot 10 + (-0.49) \cdot 2 + 0 + (-0.02) \cdot 3) \\
 &= \Phi(0.31) \\
 &= 0.6217
 \end{aligned}$$



→ I added this question to help you review Q2 in Discussion 3.

Problem 3. Using data from a private school voucher program, also known as a school choice program, we estimated the following regression. *choiceyrs* represents the number of years a student attended a choice school from 1991 to 1994. Since it is a categorical variable that takes the values 0, 1, 2, 3, and 4, we generated binary variables *choiceyrs1*, *choiceyrs2*, *choiceyrs3*, and *choiceyrs4*.

```
. reg mathscore choiceyrs1 choiceyrs2 choiceyrs3 choiceyrs4 black hispanic female
```

Source	SS	df	MS	Number of obs	=	990
Model	37872.1524	7	5410.30749	F(7, 982)	=	13.52
Residual	392958.98	982	400.161894	Prob > F	=	0.0000
				R-squared	=	0.0879
				Adj R-squared	=	0.0814
Total	430831.132	989	435.622985	Root MSE	=	20.004

mathscore	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
choiceyrs1	.5102603	1.735531	0.29	0.769	-2.895516	3.916037
choiceyrs2	-1.314817	2.098575	-0.63	0.531	-5.433025	2.803392
choiceyrs3	-2.766788	2.198078	-1.26	0.208	-7.080257	1.546681
choiceyrs4	-.617641	2.856311	-0.22	0.829	-6.222817	4.987535
black	-16.19045	1.849133	-8.76	0.000	-19.81916	-12.56174
hispanic	-13.37447	2.40842	-5.55	0.000	-18.10072	-8.648231
female	1.370047	1.277174	1.07	0.284	-1.136258	3.876351
_cons	57.04585	1.660486	34.35	0.000	53.78734	60.30436

- (a) Interpret the coefficient on *choiceyrs1* (omitted baseline = *choiceyrs* 0)

Students who attended a choice school for one year

scored 0.51 points higher in math than those who never attended, all else equal.

- (b) How would you rewrite a regression model to examine whether math scores differ between students who have ever attended a choice school and those who have never attended?

baseline

① We should generate a new dummy variable which takes the value of 1 if a student attended a choice school, and 0 otherwise.

→ $choice = \begin{cases} 1 & \text{if a student attended a choice school} \\ 0 & \text{have never attended} \end{cases}$

② We should run the following model:

$$mathscore = \beta_0 + \beta_1 choice + \beta_2 black + \beta_3 hispanic + \beta_4 female + u$$

difference in math score btw students who attended a choice school & tho who did not.

Problem 4. The following regression results are based on a dataset of working men from 1991, with *abil* representing cognitive ability and *educ* representing years of education.

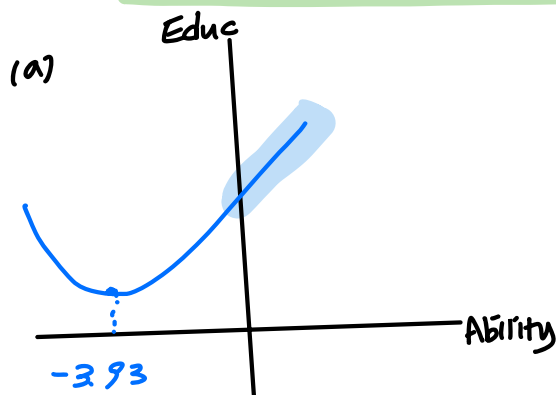
. reg educ motheduc fatheduc abil abil2

Source	SS	df	MS	Number of obs	=	1,230
				F(4, 1225)	=	244.91
Model	3027.03706	4	756.759264	Prob > F	=	0.0000
Residual	3785.24262	1,225	3.08999397	R-squared	=	0.4444
				Adj R-squared	=	0.4425
Total	6812.27967	1,229	5.54294522	Root MSE	=	1.7578

educ	Coefficient	Std. err.	t	P> t	[95% conf. interval]
motheduc	.1901261	.0280957	6.77	0.000	.1350051 .2452472
fatheduc	.1089387	.0196014	5.56	0.000	.0704827 .1473946
abil	.4014624	.0302875	13.26	0.000	.3420413 .4608835
abil2	.050599	.0083039	6.09	0.000	.0343076 .0668905
_cons	8.240226	.2874099	28.67	0.000	7.676356 8.804097

$$\text{Educ} = \beta_0 + \beta_1 \text{abil} + \beta_2 \text{abil}^2 + \beta_3 \text{mothu} + \beta_4 \text{fedu} + u$$

- (a) Draw the relationship between cognitive ability (*abil*) and years of education (*educ*).
- (b) Find the turning point.
- (c) Conduct a significance test at the 5% level to evaluate whether the relationship between cognitive ability and years of education is linear.



(b) Turning pt: slope = 0

$$\frac{\partial \text{Educ}}{\partial \text{Abil}} = 0$$

$$\Rightarrow \beta_1 + 2\beta_2 \text{Abil} = 0$$

$$\text{Abil}^* = -\frac{\beta_1}{2\beta_2} = -\frac{0.401}{2(0.05)} = -3.93$$

not meaningful
b/c ability is not negative
(can't be)

(c) ① $H_0: \beta_2 = 0$
 $H_a: \beta_2 \neq 0$

② sig lv $\alpha = 0.05$

③ test stat = $\frac{\hat{\beta}_2 - \beta_{2,0}}{\text{se}(\hat{\beta}_2)} = \frac{\hat{\beta}_2}{\text{se}(\hat{\beta}_2)}$
 $= \frac{0.050599}{0.0083039} = 6.09$

④ p-value = $P(|t| > |\text{test-stat}|)$

$$= P(|t| > 6.09)$$

$$\approx 2 \times P(Z < -6.09)$$

$$= 2 \times 0 = 0$$

⑤ Conclusion

p-value of 0 is smaller than the sig. lv 0.05.

Therefore, we reject H_0 that $\beta_2 = 0$
 $\therefore$ Educ & Ability have a nonlinear rel.

Problem 5. Using the same dataset, we run a different regression to examine the effect of number of siblings and college tuition. *sibs* is the number of siblings and *tuit17* is college tuition at age 17.

→ `gen sib3 = sibs >= 3` ← dummy $sib3 = \begin{cases} 1 & \text{if \#siblings} \geq 3 \\ 0 & \text{o/w} \end{cases}$

→ `reg educ c.tuit17##sib3` = $tuit17 \text{ sib3} \quad tuit17 \times sib3$

Source	SS	df	MS	Number of obs	=	1,230
Model	305.529838	3	101.843279	F(3, 1226)	=	19.19
Residual	6506.74984	1,226	5.30730003	Prob > F	=	0.0000
				R-squared	=	0.0448
				Adj R-squared	=	0.0425
Total	6812.27967	1,229	5.54294522	Root MSE	=	2.3038

	educ	Coefficient	Std. err.	t	P> t	[95% conf. interval]
β_1	tuit17	.0453057	.0224781	2.02	0.044	.0012059 .0894056
β_2	1.sib3	-.7144889	.3028995	-2.36	0.018	-1.308748 -.1202302
β_3	sib3#c.tuit17	-.0297482	.0318676	-0.93	0.351	-.0922694 .0327729
β_0	_cons	13.16868	.2116824	62.21	0.000	12.75338 13.58398

$$Educ = \beta_0 + \beta_1 tuit17 + \beta_2 sib3 + \beta_3 (tuit17 \times sib3) + u$$

- (a) Draw the relationship between years of education and college tuition at age 17 (*educ*) for those who have ~~more than three~~ ^{or more} siblings and those who have fewer ~~than three~~. Label the slope and intercept terms. (Ignore the statistical significance of the estimates for now.)

- (b) Can we conclude that the effect of college tuition at age 17 on years of education differs between those with more ~~than three~~ ^{three or} siblings and those with ~~three or~~ fewer? Explain. $\Rightarrow \beta_3$

- (c) Redraw the relationship if necessary, based on your conclusion from (b).

3+ siblings

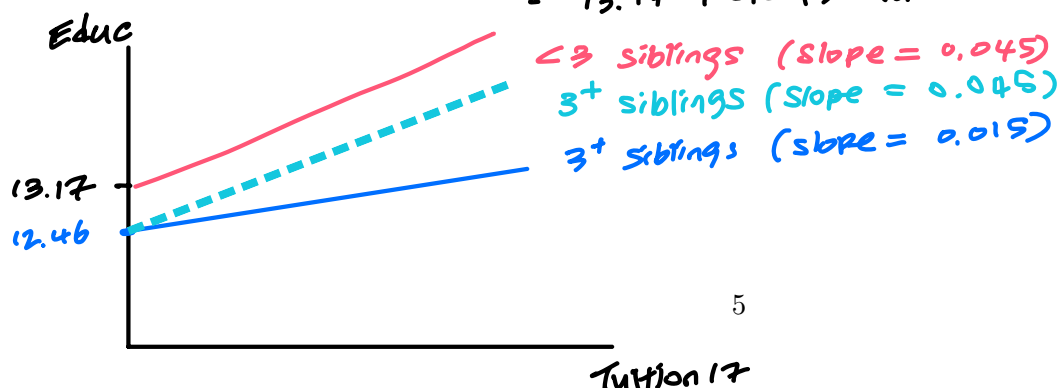
$$E[educ | tuit17, sib3=1] = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \cdot tuit17$$

$$= 12.46 + 0.015 tuit17 \quad \checkmark$$

<3 siblings

$$E[educ | tuit17, sib3=0] = \beta_0 + \beta_1 tuit17$$

$$= 13.17 + 0.045 tuit17$$



No, the estimated coeff on the interaction term is not statistically sig. at the 5% lv.

$\Rightarrow$ 3+ sib group and <3 sib group

have the same slope.

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

Problem 1. You decided to examine how macroeconomic conditions affect drug overdose deaths. Suppose you ran the following simple regression using county-level data and obtained the results. The dependent variable, Overdose Death Rate, represents the number of deaths per 100,000 people, and the variable of interest, Unemployment Rate, is measured in percentage.

$$\text{Overdose Death Rate}_i = \beta_0 + \beta_1 \text{Unemployment Rate}_i + u_i$$

- (a) Interpret the coefficients. (Assume that $\beta_0, \beta_1 > 0$.)

β_0 : Expected odd rate when ue rate is zero.

β_1 : if ue rate increases by 1pp, the expected odd rate increases by β_1 .

- (b) Do you think the regression suffers from omitted variable bias? If so, give one example of a confounder and predict the direction of the omitted variable bias due to the confounder.

Poverty rate

True : odd rate = $\beta_0 + \beta_1 \text{UE rate} + \beta_2 \text{Poverty rate} + u_i$

Aux : Pov rate = $\delta_0 + \delta_1 \text{UE rate}$

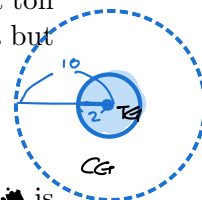
Bias = $\beta_2 \cdot \delta_1 > 0$ $\leftarrow$ explain $\Rightarrow$ An upward bias

- (c) Hollingsworth et al. (2017) use a panel data set and run the same regression model but with county and year fixed effects. According to their result, the estimated coefficient of the Unemployment Rate is (0.22) with the standard error of 0.05. Interpret the result.

if a county's UE rate increases by 1pp.

the odd rate rises by 0.22, controlling for the year FEs.

Problem 2. Currie and Walker (2011) examine the impact of introducing E-ZPass, an electronic toll collection system that reduced traffic congestion at toll plazas and, in turn, lowered pollution levels. They assess how this change affected infant health by comparing mothers living near toll plazas. In their DD design, the treatment group consists of mothers residing within 2 km of a toll plaza, while the control group includes those living between 2 km and 10 km from a toll plaza but still near a highway.



$$Y_{it} = \beta_0 + \beta_1 \text{E-ZPass}_{it} + \beta_2 \text{Close}_{it} + \beta_3 (\text{E-ZPass}_{it} \times \text{Close}_{it}) + u_{it}$$

E-ZPass is an indicator equal to one if the closest toll plaza has implemented E-ZPass; close is an indicator equal to one if the mother lived within 2 km of a toll plaza.

Fill in the table in terms of $\beta_0, \beta_1, \beta_2$, and β_3

$EZ = \phi$

$EZ = 1$

	$close = 1$ <2 km E-ZPass	$close = \phi$ >2 km and <10 km E-ZPass
Low birth weight (before)	$E(y c=1, EZ=\phi) = \beta_0 + \beta_2$	$E(y c=\phi, EZ=\phi) = \beta_0$
Low birth weight (after)	$E(y c=1, EZ=1) = \beta_0 + \beta_1 + \beta_2 + \beta_3$	$E(y c=\phi, EZ=1) = \beta_0 + \beta_1$

The summary statistics show the following:

	<2 km E-ZPass	>2 km and <10 km E-ZPass
Low birth weight (before)	0.082	0.089
Low birth weight (after)	0.078	0.092
Difference, after-before	$0.078 - 0.082 = -0.004$	$0.092 - 0.089 = 0.003$

$\Rightarrow \beta_{DD}$

Source: Currie and Walker (2011) Table 1.

$= -0.004$
 $- 0.003$

(a) Find the causal effect of E-ZPass on low birth weight.

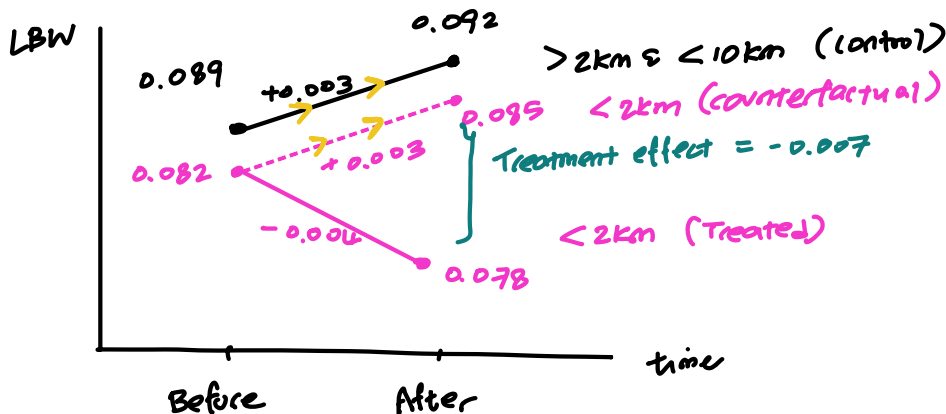
$$E(y|c=1, EZ=1) - E(y|c=1, EZ=\phi) = -0.004$$

$$E(y|c=\phi, EZ=1) - E(y|c=\phi, EZ=\phi) = 0.003$$

$$\therefore \text{Causal effect} = \Delta_{<2km} - \Delta_{>2km \& <10km} = -0.004 - 0.003 = -0.007$$

(b) In the space below, draw the following:

- Low birth weight before and after for mothers living within 2 km of an E-ZPass.
- Low birth weight before and after for mothers living between 2 km and 10 km of an E-ZPass.
- Low birth weight for mothers living within 2 km of an E-ZPass in the absence of E-ZPass adoption.
- The treatment effect.



Problem 3. In the U.S., individuals become eligible for Medicare when they turn 65. Many economists have used this age threshold to study the effect of health insurance on the utilization of health care services. Suppose we have monthly insurance claims data for individuals from 12 months before to 12 months after their 65th birthday.

(a) Fill out the following:

- Treatment: *being eligible for Medicare*
- Outcome: *use of healthcare services*
- Running variable: *Age*
- Cutoff: *65*
- Treated individuals: *Age ≥ 65*
- Control individuals: *Age < 65*

(b) Write out an equation to estimate an RD design with different slopes before and after the cutoff. Draw an RD graph assuming that all the coefficients are positive. Label the following:

- The treatment and control groups, the x- and y-axes, and the cutoff
- $E[Y|\text{Treat} = 1] = (\beta_0 + \beta_1) + (\beta_2 + \beta_3)(\text{Age} - 65)$
- $E[Y|\text{Treat} = 0] = \beta_0 + \beta_2(\text{Age} - 65)$
- $E[Y|\text{Treat} = 1]$ at the cutoff $\beta_0 + \beta_1$
- $E[Y|\text{Treat} = 0]$ at the cutoff β_0
- The treatment effect β_1

$$y_i = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \overbrace{(\text{Age} - 65)}^{\text{distance}} + \beta_3 \overbrace{[\text{Treat} \times (\text{Age} - 65)]}^{\text{distance}} + u_i$$

